Apache DataSketches

Approximate answers from compact aggregate state.

Install & load

```
INSTALL datasketches FROM community;
LOAD datasketches;
```

01 At a glance

Build sketches over rows, persist their typed state, and query estimates later. Select a family for distinct counts, quantiles or frequent items; merge states to summarize partitions.

Rows → Sketch state → Merge / estimate

02 Core SQL surface

`datasketch_hll / datasketch_cpc`

Approximate distinct-count sketches; query with _estimate.

`datasketch_theta`

Distinct-count sketch with union, intersection and difference.

`datasketch_kll`

Quantile sketch; query with _quantile, _rank, _cdf or _pmf.

`datasketch_quantiles / datasketch_req`

Alternative quantile families with different error behavior.

`datasketch_tdigest`

Compact distribution summary for quantile estimates.

`datasketch_frequent_items`

Track frequent strings or numbers and their estimated counts.

`datasketch_hll_union(lg_k, sketch)`

Aggregate compatible HLL states before estimating the union.

03 Keep in mind

Merge states, not estimates

Adding daily distinct estimates double-counts people present on multiple days. Merge compatible sketches first, then ask for the final estimate.

04 Count distinct values

```
SELECT datasketch_hll_estimate(
  datasketch_hll(12, i % 1000)
) AS distinct_values
FROM range(10000) AS t(i);
```

An approximate count of 1,000 distinct values from 10,000 input rows.

05 Estimate p50 and p95

```
WITH agg AS (
  SELECT datasketch_kll(200, i::DOUBLE) AS s
  FROM range(10000) AS t(i)
)
SELECT datasketch_kll_quantile(s, 0.50, true)
         AS p50,
       datasketch_kll_quantile(s, 0.95, true)
         AS p95
FROM agg;
```

The trailing true requests inclusive search. Input type determines sketch type.

06 Inspect retained state

```
SELECT datasketch_kll_n(
  datasketch_kll(200, i::DOUBLE)
) AS input_count
FROM range(10000) AS t(i);
```

The sketch tracks the number of input items, independently of its retained sample.

07 In practice

Match accuracy to the workload

lg_k controls HLL size; k controls KLL capacity. Larger settings use more memory. Quantile rank error is not an absolute error bound on the returned value.

Full docs & examples ↗ Source ↗